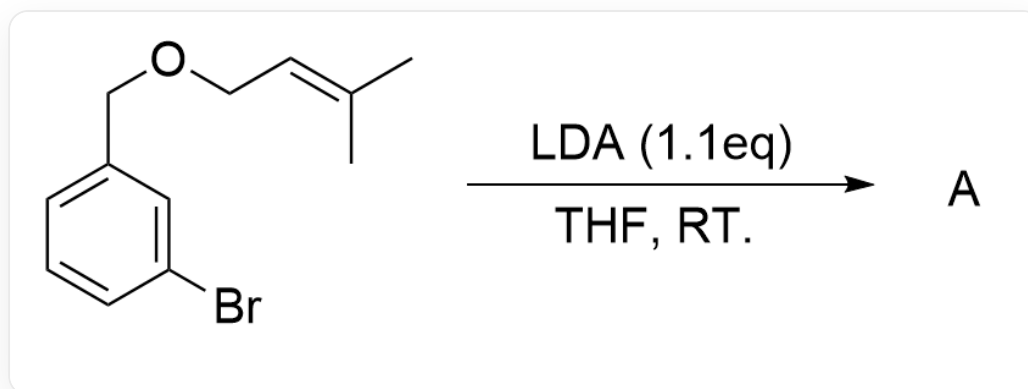


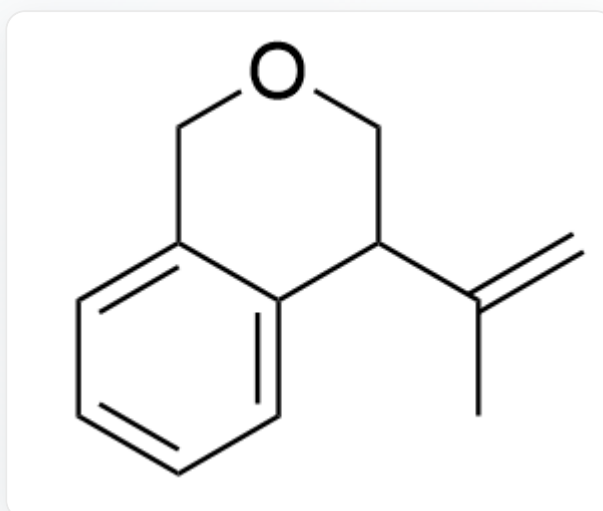
Question



BrC1=CC(COC/C=C(C)C)=CC=C1 > *LDA* (1.1eq), *THF*, R.T. > [A], where *LDA* is lithium diisopropylamide and *THF* is tetrahydrofuran

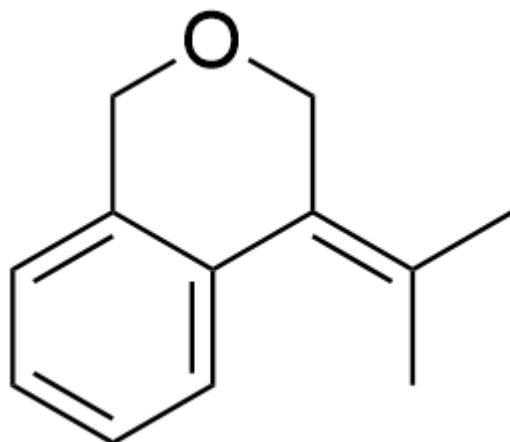
Please select the main product A of this reaction

A.



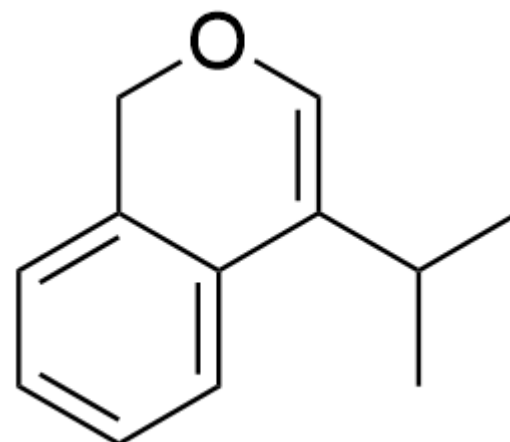
CC(C1COCC2=CC=CC=C21)=C

B.



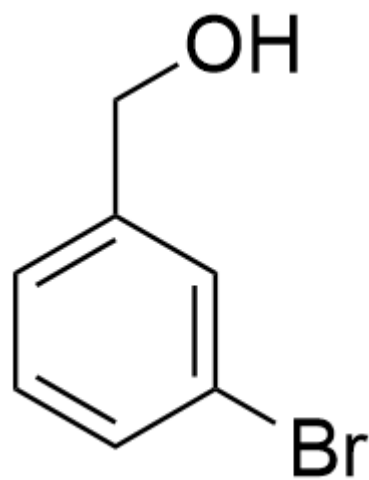
C/C(C)=C1COCC2=CC=CC=C2/1

C.



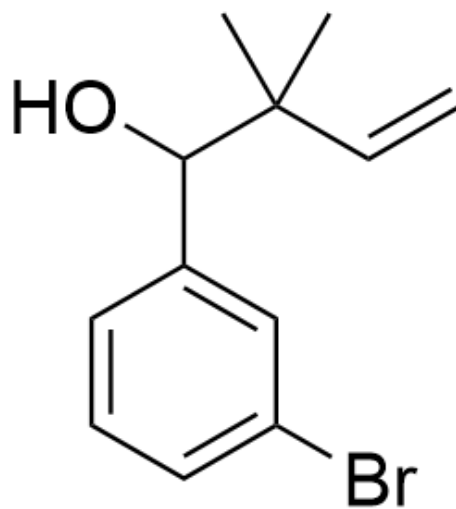
CC(C)C1=COCC2=CC=CC=C21

D.



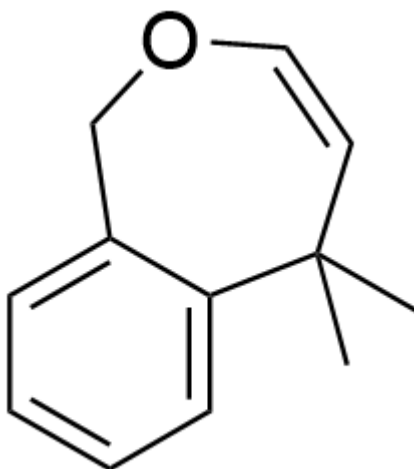
BrC1=CC(CO)=CC=C1

E.



BrC1=CC=CC(C(C(C)(C=C)C)O)=C1

F.



CC1(C)C2=CC=CC=C2COC=C1.C.C

Answer

Correct Answer: **E**

Detailed Explanation

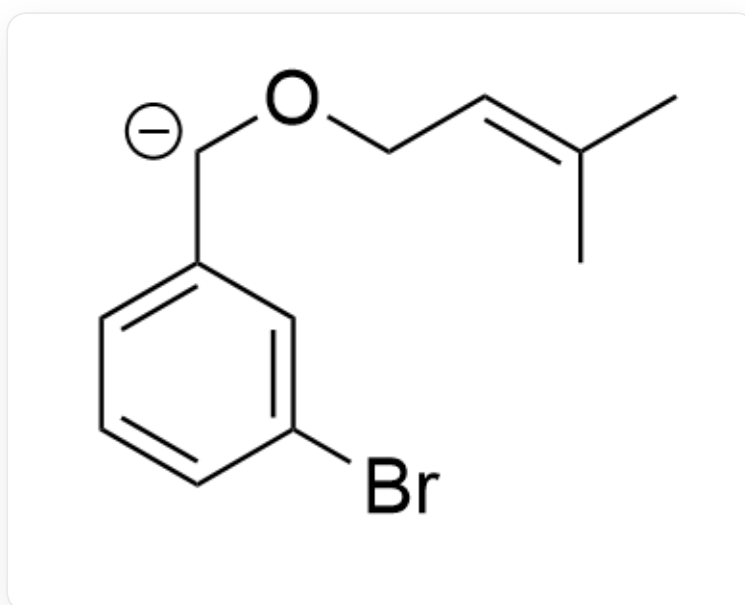
Benzene rings without strong electron-withdrawing groups are difficult to form benzyne intermediates.

CHECKPOINT

1 PTS

Benzene rings without strong electron-withdrawing groups are difficult to form benzyne intermediates

The most acidic site of the substrate is the proton at the benzylic position. Therefore, *LDA* abstracts the proton at the benzylic position, first forming a carbanion intermediate:



BrC1=CC([CH-]OC/C=C(C)C)=CC=C1

CHECKPOINT

1 PTS

LDA abstracts the proton at the benzylic position, first forming a carbanion intermediate

Then a one-step [2,3]-Wittig rearrangement occurs to give the final product E.

CHECKPOINT

1 PTS

Then a one-step [2,3]-Wittig rearrangement occurs to give the final product E